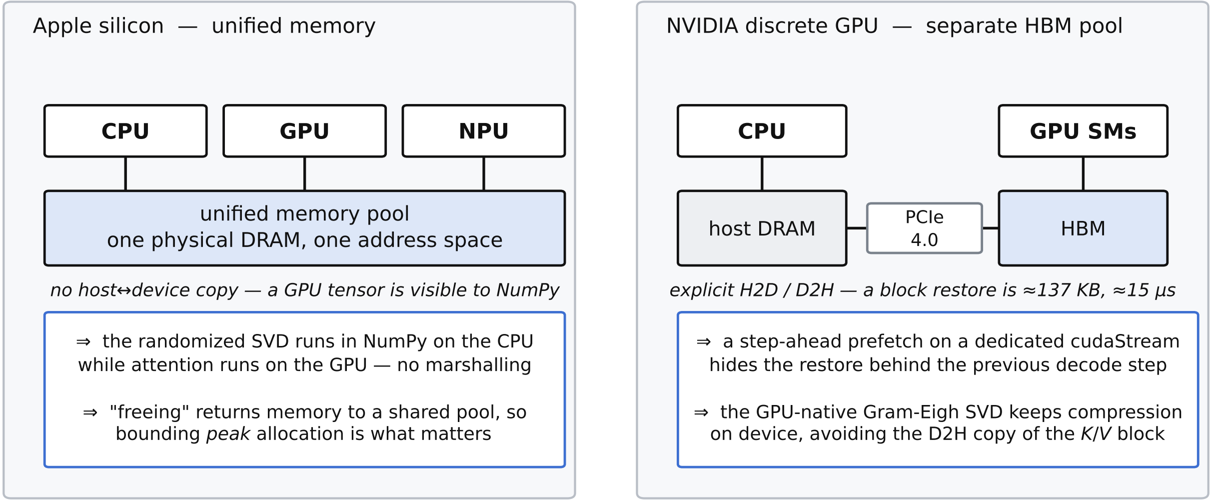
Apple silicon — unified memory
CPU
GPU
NPU
unified memory pool
one physical DRAM, one address space
no host↔device copy — a GPU tensor is visible to NumPy
⇒ the randomized SVD runs in NumPy on the CPU while attention runs on the GPU — no marshalling
⇒ "freeing" returns memory to a shared pool, so bounding *peak* allocation is what matters
NVIDIA discrete GPU — separate HBM pool
CPU
GPU SMs
host DRAM
PCIe 4.0
HBM
explicit H2D / D2H — a block restore is ≈137 KB, ≈15 µs
⇒ a step-ahead prefetch on a dedicated cudaStream hides the restore behind the previous decode step
⇒ the GPU-native Gram-Eigh SVD keeps compression on device, avoiding the D2H copy of the *K/V* block